ALI ISMAIL

Software Engineer Intern | AI Platform Engineering, LLM Observability and Evaluation

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Portfolio: aliismail293.github.io | *how-it-works walkthroughs and interactive system designs for every project*

EXPERIENCE

AI Engineering Intern | TELUS Health, AI Accelerator | Toronto, ON — May 2026 – Aug 2026

- Shipped an LLM observability platform on self-hosted Langfuse covering health, usage, cost, quality and ownership for **24 production AI applications**, **replacing six separate systems** for **~30 users** across engineering, SRE and leadership.
- Built the team's first alerting on Cloudflare Workers, firing only on state transitions with severity routing and recovery notices; **13 of 33 tracked projects** sat in constant breach, so per-breach alerts would have fired daily.
- Designed a shared health engine for answer quality, hallucination rate and P95 latency with per-project overrides, read by the UI, alerting and assistant; flagged **4 projects** below the quality threshold and **2** above the 10-second P95 limit.
- Built per-project LLM cost attribution mapping opaque Apigee project IDs to business units, plus a value-vs-cost view of active users against spend that informed **decommissioning 2 low-usage projects**.
- Built a grounded AI assistant over live telemetry using query rewriting, logical routing across four sources and metadata filtering rather than vector search, so answers carry current figures; used by **~65% of platform users**.
- Architected server-side credential brokering through a single service account so ~30 users hold no BigQuery, Langfuse or GCP credentials; extended the BigQuery ETL pipelines and separated warehouse data from Cloudflare D1 application state with cache fallback.
- Prototyped an agentic code-testing tool with multi-step tool-calling loops; diagnosed an **~8,700-line input** that overflowed the context window and caused hallucination, then designed a sub-agent running-summary memory strategy to fix it.

PROJECTS

Full walkthroughs and system designs: **aliismail293.github.io**

ragaudit | Open-source RAG evaluation and tuning tool | aliismail293.github.io/rag-tester — Sep 2026 – Oct 2026

Python, Pydantic, SQLite, Typer, Jinja2, httpx, pytest, Docker, Ollama

- Built and shipped an open-source RAG auditor that measures whether retrieval contributed to an answer, catching "lucky passes" where the model answers correctly from memory and answer-only evals score a pass.
- Demonstrated **100% answer correctness against a 30.8% retrieval hit rate** on a purpose-built corpus: **9 of 13 cases were lucky passes**, each confirmed by re-running with retrieval disabled.
- Added test-set generation from any corpus that rewrites questions under a **0.5 lexical-overlap cap** and verifies each answer key with a second, different model, so scores reflect real retrieval rather than copied wording.
- Built a tuning loop that searches pipeline configs on a dev set and scores once on a **code-locked test set split by chunk** to block leakage; a result cache re-tests only changed chunks; **272 tests** run with zero network calls.

Who's Real? | Multiplayer social-deduction game on Roblox | aliismail293.github.io — Oct 2026 – Present

Luau (strict), Rojo, ProfileStore, StyLua, Selene, Roblox Studio

- Built a server-authoritative multiplayer game on a **12-service round state machine** where clients never receive hidden roles, and validated every remote by round state, sender and rate limit to stop exploit clients.
- Isolated game rules (role rolls, votes, kill line-of-sight, win checks) into pure modules covered by **31 unit tests**; held a **60 fps server with 15 bots** and no memory growth over 6 rounds; launching Dec 2026.

EDUCATION

University of Toronto | Bachelor's Degree — Expected 2029

Minors in Computer Science and Mathematical Sciences.

Coursework 2026–27: Software Design, Theory of Computation, Data Structures and Analysis, Linear Algebra, Multivariable Calculus, Probability and Statistics.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Rust, Luau

AI and LLM: Retrieval-augmented generation (RAG), query rewriting, logical routing, metadata filtering, LLM-as-a-judge evaluation, agentic tool-calling, context engineering, local models (Llama 3.1 via Ollama), OpenAI and Gemini APIs

LLMOps and Observability: Langfuse, tracing, cost attribution, P95 latency monitoring, alerting, ablation testing, synthetic test-set generation

Data, Cloud and Tools: BigQuery, SQLite, ETL pipelines, Google Cloud Platform, Cloudflare Workers and D1, Apigee, Docker, Git, Linux, REST APIs, React, Looker, n8n